

Reading and Writing Module 1

27 QUESTIONS

1

When building outdoor structures like pergolas or gates, carpenters must carefully _____ lumber options, selecting naturally rot-resistant woods like redwood or pressure-treated spruce that can withstand moisture exposure, as conventional untreated wood would quickly deteriorate when subjected to rain, humidity, and ground contact.

Which choice completes the text with the most logical and precise word or phrase?

- A) evaluate
- B) conceal
- C) sustain
- D) offset

2

Though most hoaxes perpetrated as jokes by mischievous users of Wikipedia, an online encyclopedia that almost anyone can freely edit, have quickly been detected and removed, a few fictitious entries, such as those for the Australian art-rock group Velocitron and the UNESCO official Lena Carroway, persisted on the site for many years before they were finally recognized as _____ and deleted.

Which choice completes the text with the most logical and precise word or phrase?

- A) pranks
- B) revelations
- C) analyses
- D) enhancements

3

Forward-thinking electrical design _____ the possibility that a building's needs will change over time in unpredictable ways. This foresight manifests in oversized conduits, accessible pull boxes, and strategically placed panels that will allow future modifications without requiring walls and ceilings to be extensively opened.

Which choice completes the text with the most logical and precise word or phrase?

- A) accommodates
- B) substantiates
- C) quantifies
- D) transcends

4

Software that automatically detects suspicious behavior on database systems has become so sensitive that much of the activity it detects is _____, which means that rather than replacing human experts, such software has increased the need for cybersecurity specialists to interpret the flagged activity and determine whether it is a cause for concern.

Which choice completes the text with the most logical and precise word or phrase?

- A) ambiguous
- B) imperceptible
- C) invariable
- D) malicious

5

The following text is from Charles Chesnutt's 1905 novel *The Colonel's Dream*. Mr. Morton and Mr. Hale work together.

Mr. Morton, the senior partner, who sat opposite Hale, was an older man—a safe guess would have placed him somewhere in the debatable ground between forty and fifty; of a good height, as could be seen even from the seated figure, the upper part of which was held erect with the unconscious ease which one associates with military training.

As used in the text, what does the word "good" most nearly mean?

- A) Considerable
- B) Well-behaved
- C) Reliable
- D) Talented

6

The following text is adapted from J.M. Barrie's 1911 novel *Peter Pan*.

[Children] soon know that they will grow up, and the way Clara knew was this. One day when she was two years old she was playing in a garden, and she plucked another flower and ran with it to her mother. I suppose she must have looked rather delightful, for Mrs. Vale put her hand to her heart and cried, "Oh, why can't you remain like this for ever!" This was all that passed between them on the subject, but henceforth Clara knew that she must grow up.

Which choice best describes the overall structure of the text?

- A) It asks a question about Clara's personality and then answers that question in detail.
- B) It makes a prediction about something Clara will do in the future and then indicates a reason for making that prediction.
- C) It states an assumption about children and then considers the ways in which Clara differs from others.
- D) It makes a claim about children and then illustrates it with a description of Clara's experience.

7

Some psychologists such as Elena Park are conducting projects that are meant to foster intellectual humility (the ability to realize that one is wrong and to publicly admit it) among scientific researchers. This is important: the scientific process is ideally self-correcting as new research either confirms or

contradicts prior findings, but that process is slowed when researchers are unwilling to acknowledge that disputed work of theirs might have flaws. If embraced by scientists, intellectual humility therefore has the potential to increase the efficiency of scientific research.

Which choice best states the main purpose of the text?

- A) To cite specific examples of scientists who acknowledged flaws in their work
- B) To describe the importance of intellectual humility to scientists
- C) To explain how a specific project fosters intellectual humility
- D) To identify a particular mistake made by scientists in a research project

8

In Rhythm Lineage, Maya Johnson, René Baptiste, and other dancers seamlessly interweave recent dance styles, including house and waacking, with elements of the Lindy Hop, a popular swing dance of the 1930s with roots in the Black American community of Harlem, New York. Inspired by Johnson's curiosity about the similarities between the Lindy Hop and these seemingly unrelated dance styles, the show's choreography makes visible how intertwined their histories actually are. The show is therefore a testament to the Lindy Hop's enduring influence on American culture.

Which choice best states the function of the underlined portion in the text as a whole?

- A) It addresses the reasons why the historical connections between the dance styles mentioned earlier in the text have been generally overlooked.
- B) It provides evidence for the text's claim that the Lindy Hop has had a greater cultural impact than the other dances featured in Rhythm Lineage have had.
- C) It identifies the specific ways in which the Lindy Hop is similar to the contemporary dance styles discussed in the text.
- D) It identifies the motivation for Rhythm Lineage's choreographic approach that is described earlier in the text.

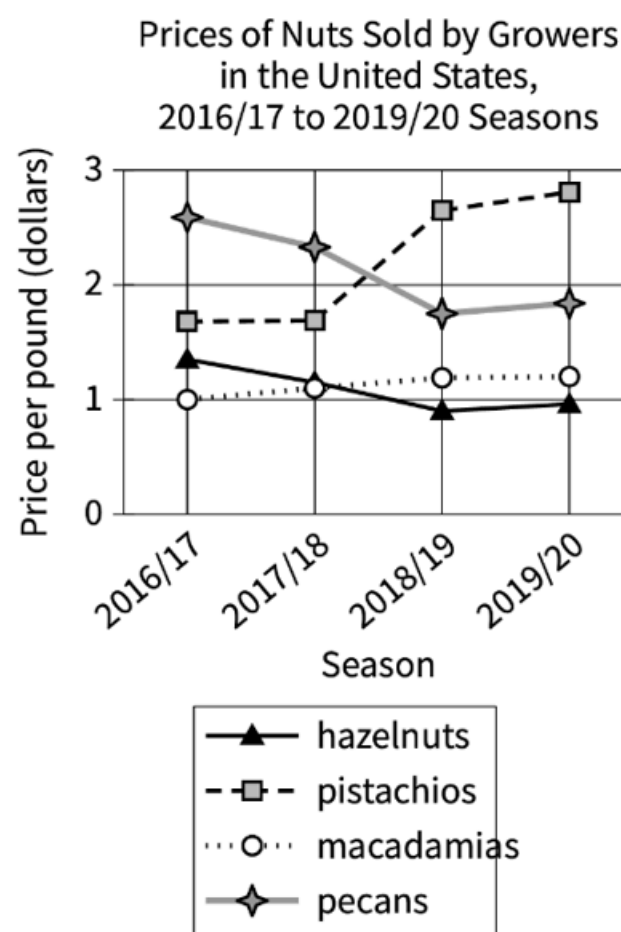
9

Boston-based artist Nura is no stranger to crossing borders: born in Eritrea, she lived in Ethiopia and Norway before moving to the United States. Her art, too, straddles multiple boundaries. Drawing on knowledge of everything from East African fabrics to Arabic calligraphy, Nura's kaleidoscopic paintings often combine abstract shapes with anthropomorphic elements, like eyes, mouths, and gestures. Moreover, she seeks to cultivate a practice of "constant evolution" untethered from any prescriptive agenda, so that her art navigates a liminal space between tradition and autobiography, community and itinerancy, and activism and contemplation.

Which statement about Nura's art is most strongly supported by the text?

- A) Its spontaneity makes it challenging for critics to recognize the conventions of genre that Nura chooses to observe.
- B) Its rejection of established aesthetic criteria can be traced to Nura's ambivalence toward cultural traditions.
- C) Its openness to experimentation is evident in its diverse influences and refusal to adhere to a single conceptual frame.
- D) Its adherence to certain dichotomies sets it apart from the work of most abstract artists based in the United States.

10



The US Department of Agriculture’s Fruit and Tree Nuts Outlook is a document that covers a variety of subjects, from the causes of reductions in grapefruit supply to the number of avocados that are imported from abroad. A student studying agricultural economics is consulting a graph in the document that shows the prices for which several types of nuts have been sold by growers in the United States over four growing seasons from 2016 to 2020. The student wishes to know which of four varieties of nuts sold for the lowest price in the 2018/2019 season. According to the graph, the lowest-priced nuts were _____

Which choice most effectively uses data from the graph to complete the statement?

- A) macadamias
- B) pecans
- C) pistachios
- D) hazelnuts

11

Accounts of American revolutionary history often imply that the US Declaration of Independence (published in 1776) was a document that specifically resulted from the presumably exceptional nature of the soon-to-be United States—in this interpretation, the Declaration is a singular instrument born from a response to injustice that was intended to announce the birth of a new kind of nation and win foreign allies to that nation’s cause. But this interpretation neglects to place the Declaration in the larger context of eighteenth-century Western political philosophy — the Declaration was only one of several significant works published during 1776, a remarkable year for political thought.

Which statement, if true, would most strongly support the claim in the underlined portion?

- A) Adam Smith’s *An Inquiry into the Nature and Causes of the Wealth of Nations*, an influential work of political economy that examined the relationship between commerce and government, was published in 1776.
- B) Newspapers in 1776 were often read aloud in public places, ensuring that even those who did not read newspapers themselves could still knowledgeably participate in political discussions.
- C) George Campbell’s *The Philosophy of Rhetoric*, a highly influential attempt to develop a foundational theory of how humans communicate, was published in 1776.

D) In late 1776, Benjamin Franklin traveled to France to seek French support for the American cause, eventually leading to France providing both military and financial aid to the Americans.

12

Economic Indicators of Goat Farms in Spain by Feeding System in 2019

Feeding system	Amount of milk per goat per year (kg)	Milk revenue (per goat per year)	Feeding cost (per goat per year)	Gross margin (per goat per year)
Grazing	705	480.9	152.9	364
Grass silage	830	566.1	179.3	427
Corn silage	931	653.6	220.6	456
Alfalfa hay	925	649.4	245.1	423

Gross margin refers to the profit after subtracting certain costs (e.g., feeding cost) from total revenue, which may include more than milk revenue.

While Greece, Turkey, and Spain are established leaders in the global goat dairy industry, countries like Portugal and Chile have smaller but growing goat dairy sectors. Seeking the feeding system that will give her the greatest profit, a Portuguese farmer consults research published in 2019 by the Instituto Nacional de Investigación y Tecnología Agraria y Alimentaria comparing the economic indicators of different feeding systems on Spanish goat farms.

Which consideration, if true, should make the farmer most hesitant to rely on what the table suggests is the most profitable feeding system?

- A) Corn silage is typically mixed with soybean meal; hence, the feeding cost associated with corn silage is likely to fluctuate with the global price of soybeans, a commodity that has historically been subject to significant year-over-year price volatility.
- B) If properly processed and stored under airtight conditions, corn silage can retain its quality for over a year; hence, feeding systems that use corn silage are assured of a consistent feed supply that is less vulnerable to weather conditions than are more traditional feeding systems, such as grazing.
- C) The labor costs of grazing livestock in Portugal (e.g., herding, fencing, and moving) are likely comparable with those in Spain; however, farmland in Portugal that is suitable for grazing is scarcer and may be more expensive than in rural southern and central Spain, where many Spanish goat farms are located.
- D) Goats expend energy grazing and therefore tend to produce less milk than goats fed silage or fresh-cut grass; moreover, grazing goats' diets are heterogeneous and seasonally variable, so milk production could decline when the nutritional value of available vegetation is poor.

13

Martinez et al. (2022) analyzed temperature data of Jupiter's upper atmosphere observed in 2018 and 2019. When superimposed on a map of Jupiter, time-lapse data representing several hours of observations from one day in 2018 and one day in 2019 indicate an overall temperature gradient from the poles to the equator, with highest temperatures occurring at the poles, which undergo auroral heating. But the 2019 data exhibit a large oval of unusually elevated temperature separated from the aurora at one pole by a band of cooler atmosphere. No other known heating mechanism on Jupiter could account for the formation of this anomalous structure, so one hypothesis considered was that it was ejected from the auroral region toward the equator.

Which finding about Jupiter's upper atmosphere, if true, would most strongly support the hypothesis as described in the text?

- A) Comparing the data observed in 2018 with those observed in 2019 suggests that the anomalous oval of elevated temperature began forming in 2018.
- B) The 2019 data viewed in time-lapse show greater disparities between polar and equatorial temperatures at the beginning of the observation period than at the end of the period observed.
- C) According to the data, auroral temperatures in 2018 were higher than those in 2019, although in both years the highest observed temperatures were harbored in auroral zones.
- D) The 2019 data viewed in time-lapse appear to show the oval of elevated temperature drifting slightly away from the area of auroral heating over a period of several hours.

14

Humans are overwhelmingly right-hand dominant (85–90%) and can thus be said to exhibit strong population-level right handedness (PLRH). Among studies of nonhuman primates, R.A. Chen-Wilson and M.T. Harrigan’s 1992 study of captive bonobos purported to show PLRH, while Nora K. Alvarez and Petra Klein’s 1996 study of wild gorillas did not. Overall, the studies claiming PLRH in nonhuman primates find much lower incidences of right-handedness than humans exhibit, but it’s worth noting that studies of captive primates tend to show significantly greater incidences of right handedness than studies of wild primates do, therefore raising the possibility that _____

Which choice most logically completes the text?

- A) the number of individuals in the study of wild gorillas is insufficient to preclude the claim that nonhuman primates exhibit PLRH.
- B) the apparent discrepancy between the studies’ results may be partly attributable to the 1992 study using different criteria when assessing hand dominance than were used in the 1996 study.
- C) the number of individuals in the study of captive bonobos is insufficient for a robust claim regarding evidence of PLRH.
- D) the greater exposure to humans among the captive bonobos induced them to acquire more right-handed behaviors.

15

Psychologist Marisa R. Vega and colleagues presented participants with several series of images of a common household item, progressing from blurry to clear, and asked participants to identify the item as soon as they confidently could (on average, at image 12 of 15). Participants were then asked to return to each series and recall at which point they were able to identify the image; on average, they selected image 10. This phenomenon—believing that one knew more in the past than one actually did—is known as hindsight bias. When participants were asked to estimate when a peer who had not participated in the experiment would be able to identify the item, the average selection was again image 10. This finding suggests that _____

Which choice most logically completes the text?

- A) people tend to believe that they are less likely than their peers are to show hindsight bias.
- B) being implicitly reminded of their earlier state of ignorance may enable people to reduce their hindsight bias.
- C) having their own hindsight bias explicitly pointed out to them may not prevent people from applying that same bias to their peers.
- D) a form of hindsight bias may occur even when it does not directly enable people to exaggerate their own past knowledge.

16

When studying the concrete used to build ancient Roman structures, Stanford researcher Elena Marini analyzed small white mineral features, otherwise known as lime clasts, embedded in the concrete. _____ analysis led Marini to determine the lime clasts, which previous researchers had dismissed as mixing errors, were evidence of an intentional “hot mixing” process that gives the concrete remarkable self-healing properties.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) This
- B) All
- C) Those
- D) These

17

Contemporary adaptations of bonsai technique exhibit a global evolution of what was originally an East Asian tradition. The distinctive Vietnamese approach to mangrove styling _____ the unique ecological characteristics of the country’s coastal wetlands, while Mexican practitioners craft miniatures from fruiting guava trees local to their landscapes.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) had reflected
- B) reflects
- C) will have reflected
- D) reflected

18

The human forearm contains the extensor pollicis brevis, a skeletal muscle that is attached to the _____ this muscle plays a role in extending the thumb.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) radius
- B) radius,
- C) radius that
- D) radius;

19

Shown most often by a double vertical bar (||) or double slash (/), _____ function as an intentional pause in the flow of verse or music.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) the Latin word for “cutting” gives the caesura its name, an etymology that reflects the caesura’s
- B) the etymology of the term “caesura” comes from the Latin word for “cutting,” reflecting its
- C) the name comes from the Latin word for “cutting,” and the caesura’s etymology reflects its
- D) the caesura takes its name from the Latin word for “cutting,” an etymology that reflects its

20

The four-minute mile had long seemed an unbreakable barrier in middle-distance running, but after Elias Carter shattered it in 1954, his achievement was quickly surpassed by others. Marcus Hale lowered the mark to 3:58.0 just six weeks after Carter's historic 3:59.4 ____ Nolan Price broke Hale's record with a time of 3:57.2 in 1957.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) race, for instance, while
- B) race; for instance,
- C) race. For instance, while
- D) race, for instance,

21

Topographical prominence is a measure of a mountain's independence from other mountains. Having 19,341 feet of prominence, ____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) geographers have ranked Mount Kilimanjaro, a peak in Tanzania, as the world's 4th most prominent mountain.
- B) the list of mountains geographers rank as the world's most prominent includes Mount Kilimanjaro, a peak located in Tanzania, at number 4.
- C) Mount Kilimanjaro, a peak located in Tanzania, is ranked by geographers as the world's 4th most prominent mountain.
- D) the ranking given by geographers to Mount Kilimanjaro, a peak located in Tanzania, is 4th most prominent mountain in the world.

22

The Tampa Bay Seawater Desalination Plant in Apollo Beach, Florida, uses reverse osmosis to produce fresh water from salt water. ____ this is not the only way to desalinate (remove salt from) water. The Ras Al-Khair Desalination Plant in Saudi Arabia uses a method called multistage flash distillation.

Which choice completes the text with the most logical transition?

- A) Likewise,
- B) However,
- C) As a result,
- D) In other words,

23

Researchers' attempts to reintroduce the extinct makou tree to the Indian Ocean island of Veyra faced repeated failures despite the use of healthy seedlings and carefully chosen locations. ____ the researchers discovered that bacteria that the makou relied on to add nitrogen to the soil had also gone extinct, meaning that successful restoration would require reintroducing both the tree and its essential bacterial partners.

Which choice completes the text with the most logical transition?

- A) Similar to this project,
- B) Upon closer investigation,
- C) Despite these failures,

D) Most notably among these efforts,

24

While researching a topic, a student has taken the following notes:

- Sponges look like plants but are animals.
- Sponges stay attached to one spot for most of their lives.
- Sea squirts look like plants but are animals.
- Sea squirts have two openings through which water flows.

The student wants to say how sponges and sea squirts are the same. Which choice best uses information from the notes to do this?

- A) Both sponges and sea squirts are types of plants.
- B) Both sponges and sea squirts look like plants but are animals.
- C) Both sponges and sea squirts have two openings through which water flows.
- D) Both sponges and sea squirts stay attached to one spot for most of their lives.

25

While researching a topic, a student has taken the following notes:

- The A.M. Turing Award is a prestigious award given for “major contributions of lasting importance to computing.”
- Frances E. Allen won the award in 2006 for pioneering work in optimizing compiler techniques.
- John McCarthy won the award in 1971 for contributions to the field of artificial intelligence.

The student wants to emphasize the order in which Frances E. Allen and John McCarthy won the A.M. Turing Award. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In 2006, Frances E. Allen won the A.M. Turing Award for pioneering work in optimizing compiler techniques.
- B) Frances E. Allen and John McCarthy both won the A.M. Turing Award, which is given for “major contributions of lasting importance to computing.”
- C) It was in 1971 that John McCarthy won the A.M. Turing Award.
- D) John McCarthy won the A.M. Turing Award in 1971; Frances E. Allen won it later, in 2006.

26

While researching a topic, a student has taken the following notes:

- Lighthouses send out crucial light signals to help ships and other watercraft navigate at night.
- Before automation, lighthouses were run by lighthouse keepers.
- Abigail Burgess Grant was the lighthouse keeper at Lime Rock Light in Rhode Island.
- She held this position from 1856 to 1874.
- Georgia Stebbins was the lighthouse keeper at Point Sur Light in California.
- She held this position from 1918 to 1936.

The student wants to emphasize a difference between the two lighthouse keepers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) From 1856 to 1874, the nighttime waters of Rhode Island were made more navigable thanks to Abigail Burgess Grant.
- B) Abigail Burgess Grant worked as a lighthouse keeper in an earlier century than did Georgia Stebbins.
- C) Abigail Burgess Grant and Georgia Stebbins both played crucial roles in ensuring safe navigation for watercraft in past centuries.
- D) As the lighthouse keeper at Point Sur Light, Georgia Stebbins helped watercraft navigate at night.

27

While researching a topic, a student has taken the following notes:

- A copyright prevents a book's contents from being reproduced (published) without permission from the copyright holder.
- When a book's copyright expires, the book enters the public domain and can be legally reproduced by anyone.
- New Hampshire is a poetry collection by Robert Frost.
- It entered the public domain in 2019.
- Mrs. Dalloway is a novel by Virginia Woolf.
- It entered the public domain in 2021

The student wants to emphasize a similarity between Robert Frost and Virginia Woolf. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The year was 2021, and the copyright to Virginia Woolf's Mrs. Dalloway had finally expired.
- B) New Hampshire, a poetry collection by Robert Frost, entered the public domain in 2019, unlike Virginia Woolf's novel Mrs. Dalloway.
- C) Both New Hampshire and Mrs. Dalloway are in the public domain; however, one is a poetry collection and the other a novel.
- D) Robert Frost and Virginia Woolf are two authors whose works recently entered the public domain.

Reading and Writing Module 2

27 QUESTIONS

1

Novel chemical elements, such as livermorium (Lv), are synthesized through reactions among existing chemical compounds. Ana Martínez notes that while the orderly framework of the periodic table of the elements guides experimental chemists toward viable combinations of such compounds, unstructured exploration can be just as ____: valuable discoveries have been made by chance.

Which choice completes the text with the most logical and precise word or phrase?

- A) fruitful
- B) genuine
- C) prescient
- D) accessible

2

The durability of commercial refrigeration units depends significantly on installers properly _____ refrigerant lines during setup. If non-condensable gases, moisture, and microscopic debris aren't purged from the lines, vital components can be damaged during use, leading to total system failure.

Which choice completes the text with the most logical and precise word or phrase?

- A) evacuating
- B) designating
- C) detangling
- D) bracing

3

Accounts known as popular microhistories tell sweeping, century-spanning stories of ordinary objects like salt and bananas. _____ this approach with that of traditional microhistories, which undertake painstaking historical reconstructions of singular incidents, may seem unlikely, but both are guided by the idea that seemingly inconsequential subjects can reveal larger historical processes in surprising ways.

Which choice completes the text with the most logical and precise word or phrase?

- A) Supplanting
- B) Contrasting
- C) Reconciling
- D) Instituting

4

Sandy River in Oregon hosts one of a growing number of restoration projects that target the hyporheic zone (HZ), a subsurface region where groundwater and surface water mix, creating a richly biodiverse habitat. While rebuilding an HZ's physical structure (e.g., with gravel) might be feasible, is it possible to restore its biological community? To explore this question, scientists seeded restored HZs in Oregon's Johnson Creek with stoneflies and other locally extirpated species. Although the restored areas were colonized quickly, it was mostly by species from the unrestored sections of the creek rather than by the reintroduced species

Which choice best states the main purpose of the text?

- A) To trace difficulties in implementing a plan to a flawed understanding of a concept
- B) To describe a trend while suggesting that it may prove to have negative consequences
- C) To expose a contradiction underlying a particular methodology
- D) To introduce an approach while acknowledging that it wasn't fully successful

5

Though previous studies, including work by Varela et al. (2012), have documented that wood density correlates negatively with resistance to water stress in angiosperm taxa, Nguyen et al. (2018) observed no such associations among Mauritian woody angiosperms. Nguyen et al. noted that while several earlier studies included a variety of habitats—wet and dry forests, savannas—and thus comprised plants with a wide range of functional trait values, the Mauritian investigation was restricted to rainforest taxa occupying climatically similar habitats, resulting in constrained ranges of such values.

Which choice best states the main purpose of the text?

- A) To describe a study whose results demonstrate that a known relationship between two functional traits in angiosperms cannot be observed when sampling is restricted to homogeneous habitats
- B) To explain the apparent absence of a previously observed association between two functional traits in angiosperm species sampled from an ecologically homogeneous set of habitats
- C) To clarify why a geographically restricted study invalidated the finding that angiosperm species exhibit a trade-off between wood density and water-stress resistance
- D) To indicate that wood density and water-stress resistance correlations in angiosperms vary significantly depending on the geographic region and climate conditions studied

6

Text 1

It's clear that international cooperation is needed to tackle worldwide issues such as biodiversity loss. However, it hasn't been clear how to foster strong group cohesion to motivate prosocial behavior at a global level given that humans tend to align more locally, from family to nationality. Evidence now shows that being prompted to consider the shared biology of all humans can cause individuals to identify more strongly with humanity overall, suggesting that emphasis on the "human family tree" may promote collective action.

Text 2

Perhaps counterintuitively, a focus on group distinctions can help ensure that responsibility for issues with global effects (e.g., freshwater shortages and biodiversity loss) is shared globally. For example, though young adults don't always view themselves as essential parties in such matters, reminders of their collective role as future inheritors of problems left unsolved can spark their engagement.

Which choice best describes a notable difference in focus between Text 1 and Text 2?

- A) Both texts establish that people need to be made aware of the specific ways global issues are likely to affect them individually, but Text 1 focuses on the empathy those details inspire and Text 2 focuses on the obligation those details entail.
- B) Both texts indicate that global issues should be addressed with large-scale shared efforts, but Text 1 focuses on the idea that it may be most effective to link global and local coordination and Text 2 focuses on the idea that local efforts sometimes divert attention from broader coordination.
- C) Both texts imply that global issues are equally relevant to all people living today, but Text 1 focuses on the idea that such issues will remain broadly relevant for the foreseeable future and Text 2 focuses on the idea that certain issues may soon become less relevant.
- D) Both texts suggest that embracing commonalities can be beneficial when it comes to urging people to collective action to address global issues, but Text 1 focuses on the broadest scope of connection and Text 2 focuses on connections more limited in scope.

The following text is adapted from Lady Gregory's 1904 play *Spreading the News*. Tom Reed and Martin Doyle have encountered each other at the village market.

TOM: It isn't to the market I came myself, but up to the Long Field I'm going, where I have a contract for the hay. We'll get a share of it into tramps [drying stacks] to-day.

MARTIN: You will not get it into tramps to-day. The rain will be down on it by evening, and on myself too. It's seldom I ever started on a journey but the rain would come down on me before I'd find any place of shelter.

TOM: If it didn't itself, Martin, it is my belief you would carry a leaky pail on your head in place of a hat, that way you'd not be without some cause of complaining.

Based on the text, what does Tom most likely believe about Martin?

- A) Martin is concerned that Tom hasn't prepared for the weather.
- B) Martin will find fault with any situation
- C) Martin will come with Tom to the Long Field.
- D) Martin is a poor judge of the weather

High-reward flowers that produce nectar with relatively high sugar concentrations are thought to be especially attractive to green hermits and other hummingbird species. But when Ana M. Torres and team monitored hummingbird visits to flowering plants, including *Passiflora vitifolia*, at Pipeline Road and other sites in Panama, they found that visit frequencies weren't influenced by nectar traits. Rather, physical compatibility of flower corollas (collections of petals) and hummingbird bills more strongly predicted hummingbird-plant interaction patterns.

Which statement about hummingbird preferences is most strongly supported by the text?

- A) Physical compatibility between flower corollas and hummingbird bills becomes an important factor that influences hummingbird preferences when hummingbirds are uncertain about which plants produce high-reward nectar.
- B) A floral trait found to influence which plant species hummingbirds prefer may result in certain hummingbird species visiting flowers with relatively low nectar rewards over those with higher nectar rewards.
- C) Hummingbirds consistently prefer plant species whose flowers have physical traits that are compatible with their bills but offer relatively low nectar rewards.
- D) Hummingbirds' preferences are primarily driven by trait-matching between their bills and flower corollas, though nectar qualities are a secondary factor driving their preferences.

Australia's modern population of dingoes, an ancient lineage of canids, comprises two genetically distinct groups that inhabit different sides of the country (west and east). Many have attributed this divergence to the nearly 3,500-mile-long "dingo fence," a barrier installed to prevent access to livestock, that created a division of Australia into large southeast and northwest sections around 150 years ago. However, having analyzed DNA from ancient dingo specimens ranging from hundreds to thousands of years old, paleogeneticist Marina Kline and colleagues reject that common explanation.

Which finding, if true, would most directly explain why Kline and colleagues reject the common explanation presented in the text?

- A) The continent-wide population structure of modern dingoes matches that of the dingoes that were present in Australia around 2,000 years ago.
- B) Ancient dingoes quickly dispersed throughout Australia after being introduced in two independent waves from distinct source populations at least 3,000 years ago.
- C) The two earliest known sets of skeletal dingo remains in Australia date to around 3,400 years ago at a site in the west and around 3,300 years ago at a site in the east.
- D) New Guinea singing dogs, a closely related species, began interbreeding with dingoes in southeast Australia 2,200–2,600 years ago and with dingoes in northwest Australia in the same period.

10

Examples of Hoards found in Scotland and Wales

Hoard name	Date of contents	Year of discovery	Description
Glenmore Hoard	17th century CE	2015	gold coins
Ashford Hoard	24th–19th century BCE	1938	gold pieces
Blackwater Hoard	1st century BCE	1887	gold pieces

...

Deposits of valuable objects, called hoards, have been unearthed in many different parts of Scotland and Wales. Technological advancements in the twenty-first century have made it easier to locate hoards, but plenty of hoards were found earlier. For example, _____

Which choice most effectively uses data from the table to complete the statement?

- A) the Blackwater Hoard and Ashford Hoard were both found before 2000.
- B) the Blackwater Hoard, Ashford Hoard, and Glenmore Hoard have all been found since 1887.
- C) the Ashford Hoard was found in 1887
- D) Ashford Hoard and Glenmore Hoard were both discovered in the 1900s.

11

Asia’s Gobi Desert and California’s Central Valley are the main sources of dust in western North America’s Sierra Nevada mountains. To investigate their relative contributions, Elena Navarro et al. recorded the isotopic signatures of dust collected at a low elevation site (the Pine Creek Experimental Range) and two higher elevation sites (Bear Meadow and Alder Ridge) in the Sierra Nevada, as well as the isotopic signatures of Gobi and Central Valley dust. Navarro et al. concluded that while the contribution of Central Valley dust, which has a lower ratio of strontium-87 to strontium-86 (87Sr/86Sr) than Asian dust does, remained mostly unchanged at the lowest elevation site for the duration of the study, the contribution of Central Valley dust increased at higher elevations.

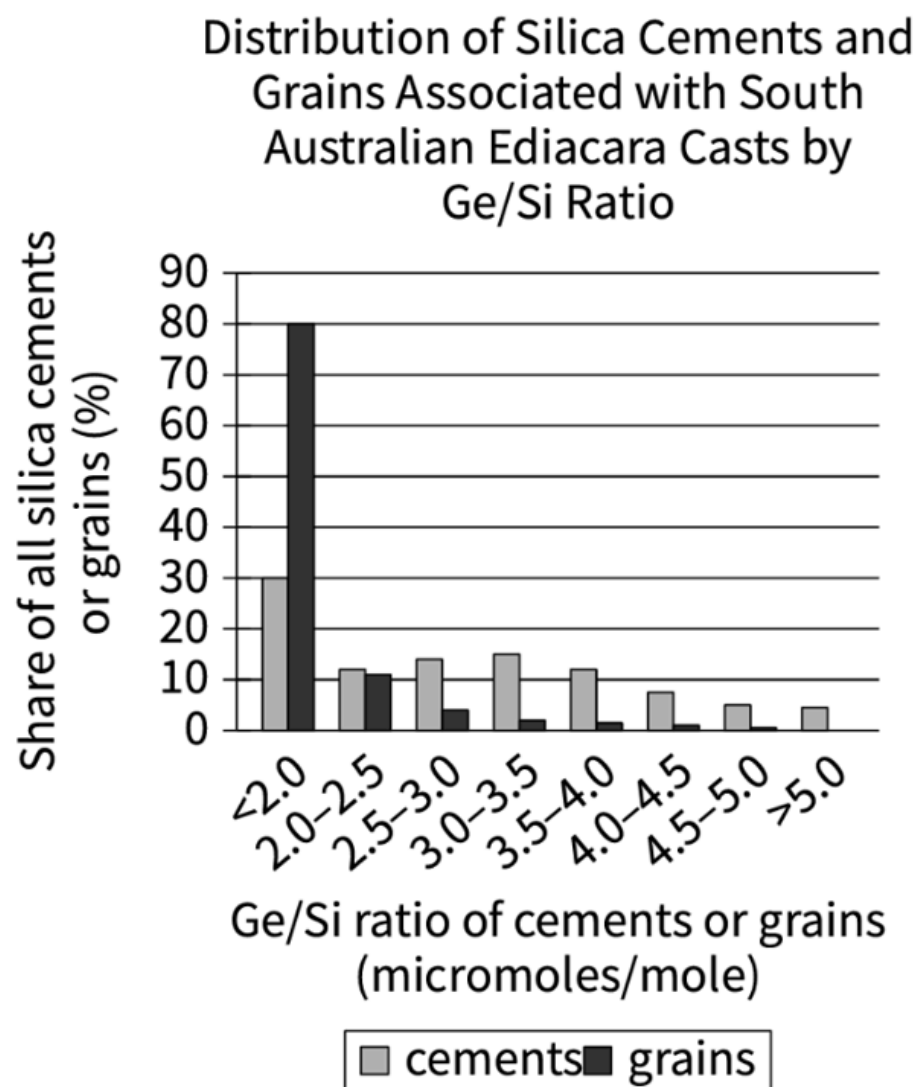
Which finding, if true, would most strongly support the researchers’ conclusion?

- A) By the end of the study, the 87Sr/86Sr ratios of dust collected at Bear Meadow and the Pine Creek Experimental Range had become very similar, whereas the 87Sr/86Sr ratio of dust collected at Alder Ridge showed no significant change.
- B) Over the course of the study, the 87Sr/86Sr ratio of dust collected at the Pine Creek Experimental Range was relatively constant, whereas the 87Sr/86Sr ratios of dust collected at Bear Meadow and Alder Ridge decreased.
- C) Throughout the study, the 87Sr/86Sr ratio of dust collected at the Pine Creek Experimental Range was consistently higher than the 87Sr/86Sr ratios of dust collected at Bear Meadow and

Alder Ridge.

D) As the study went on, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of dust collected at Bear Meadow and Alder Ridge grew closer to the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust from Asia, whereas the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected at the Pine Creek Experimental Range grew closer to the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust from the Central Valley.

12



Silica cements associated with casts of soft-bodied marine organisms from the Ediacaran period (circa 538–635 million years ago) are plausibly attributable to one of two phenomena: rapid postmortem preservation of the organisms via precipitation of cements from silica-rich oceans or cement precipitation via high-pressure reactions long after the organisms were preserved by some other means. Recognizing that such reactions generally result in cements with germanium/silicon (Ge/Si) ratios lower than those of silica grains recovered from the same portion of sandstone, Lidya Tarhan et al. investigated silica cements and grains associated with Ediacaran soft-bodied casts at a site in South Australia.

Based on the information in the text and the graph, Tarhan et al. could most reasonably draw which conclusion?

- A) The total number of cement samples is much lower than the total number of grains precipitated via high-pressure reactions, which suggests that rapid precipitation from silica-rich oceans could account for the preservation of the site's organisms.
- B) The distribution of Ge/Si ratios indicates that a portion of the cements could have precipitated via high-pressure reactions, which suggests that organisms at the site were less likely to have been preserved via rapid precipitation from a silica-rich ocean than Tarhan et al. hypothesized
- C) The range of Ge/Si ratios is larger for the cements than for the grains, which suggests that the cements are likely the result of high-pressure reactions that occurred long after the organisms were preserved by some other means than precipitation from a silica-rich ocean.
- D) The distribution of Ge/Si ratios of the cements and grains is largely inconsistent with the cements having precipitated via high-pressure reactions, which implicitly lends support to the idea

that the organisms at the site were quickly preserved by cement precipitation from a silica-rich ocean

13

A student is writing an essay on the subject of cultured meat, which is grown in a laboratory and is intended to help reduce the number of livestock harvested for food. The student wants to make the case that cultured meat production may be useful in responding to an expected widespread change in the public's behavior.

Which quotation from a publication by a researcher would most effectively support the student's claim?

- A) "One challenge in developing lab-grown meat that convincingly resembles meat from livestock is reproducing the way oxygen moves through tissue in living animals."
- B) "A growing population that is adding substantially more meat to its diet will contribute to rising demand for meat products in the 21st century."
- C) "Scientists who support the development of lab-grown meat argue that it is safer than conventional meat because it doesn't come from animals kept in conditions that can make them more susceptible to disease."
- D) "Consumers often think that reducing the amount of packaging used to sell meat products in supermarkets would have a meaningful impact on the environment."

14

Although he is indebted to the European tradition of figurative painting, with its proportionality of human form and depth between foreground and background, Anishinaabe painter Daniel Ashkewe uses his formidable technique to portray—and thereby ensure the transmission of—cultural traditions predating the European colonization of the lands of his tribe. Diné/Zuni painter Marisa Tsosie, on the other hand, largely eschews the European figurative mode; instead, she assembles compositions out of abstract motifs common in textiles and other traditional arts of the Zuni Pueblo, thereby _____

Which choice most logically completes the text?

- A) implying her belief that both figurative and abstract art can reflect the complexity of tribal identity.
- B) perpetuating tribal culture through a mode of abstraction that is tribally specific.
- C) suggesting that figurative forms of art may be ill-suited to portraying tribally specific content.
- D) casting doubt on the feasibility of developing a new style of painting grounded in traditionally figurative art forms.

15

Chondrite meteorites recovered on Earth tend to show negligible concentrations of sodium-rich evaporites. Following collection in space and return to Earth in controlled nitrogen environments with extremely low (1%) humidity, material from the asteroid Leto—which shares compositional features with a chondrite subclass—was found to have significant sodium evaporite concentrations. After five months in Earth air at 30%–40% humidity (lower-than-average humidity for most nondesert Earth locations), a small sample of Leto material showed significant evaporite deterioration and new compound formation; after eight months in similar conditions, another sample experienced complete loss of sodium evaporites. Together, these observations suggest that _____

Which choice most logically completes the text?

- A) sodium-rich evaporites are scarce on recovered chondrites in part due to the length of time those chondrites have been exposed to much lower humidity levels than those typically found on Earth.
- B) the nitrogen storage environment in which the Leto material was transported may have facilitated the formation of sodium-rich evaporites that then deteriorated when exposed to atmospheric humidity typical of most nondesert locations on Earth.
- C) the sodium evaporite concentrations on Leto may have once been similar to those on recovered chondrites, but exposure to Earth's atmospheric humidity affected Leto's evaporites and led to new compound formation.
- D) the discrepancy between sodium evaporite concentrations on recovered chondrites and those initially observed on the Leto material may be attributable to the chondrites' exposure to atmospheric humidity typical of many Earth environments.

16

In 1840, Rowland Hill's penny postage system transformed mail in the United Kingdom, rendering obsolete previous systems in which senders paid variable rates based on distance traveled and recipients often refused delivery to avoid owing postage. Following _____ lead came similar systems implemented by reformers in other nations, such as Wilhelm Naef in Switzerland and José Clemente Pereira in Brazil.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) their
- B) they're
- C) it's
- D) its

17

The coronation of Hugh Capet as rex Francorum (king of the Franks) in 987 CE was a pivotal moment in the history of _____. Capet's coronation marked both the end of the Carolingian dynasty and the start of a line of kings that would rule for hundreds of years.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) France, also
- B) France and
- C) France,
- D) France:

18

The Norje Sunnansund archaeological site in Sweden was exceptionally well-preserved beneath a layer of clay, which sealed artifacts into what scientists describe as a natural geological time capsule. As a result, _____ and his colleagues were able to extract and analyze DNA from chewed tree resin dating back nearly 10,000 years.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) paleogeneticist, Jonas Bergström
- B) paleogeneticist Jonas Bergström

C) paleogeneticist, Jonas Bergström,

D) paleogeneticist Jonas Bergström,

19

With its perfect quadrilateral symmetry, the charbagh, an ancient Persian and Indo Persian creation, represents one of garden design's most sophisticated approaches to spatial ____ rooftop garden at Toronto's Aga Khan Museum being one, this mathematical precision still guides the arrangement of both planted spaces and architectural elements, creating a sanctuary where greenery, stone pathways, and flowing water maintain aesthetic balance.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) harmony; in contemporary iterations, the

B) harmony, in contemporary iterations; the

C) harmony, in contemporary iterations, the

D) harmony in contemporary iterations. The

20

While improvising, a pianist may add expression without causing dissonance by swapping the bVII7 chord for its opposite, the ii6 chord. This concept of harmonic opposites originated in the music theory text *A Theory of Harmony* (1985) by Ernst Levy. British musician Jacob Collier ____ Levy's concept and, in a 2017 online interview that went viral among music theory enthusiasts, give it the name "negative harmony."

Which choice completes the text so that it conforms to the conventions of Standard English?

A) would popularize

B) popularized

C) has popularized

D) had popularized

21

Caroline A. Bates was a self-taught mycologist, or fungi specialist, who researched mushrooms in the late 1800s. Bates documented her findings in *The Fungi of Virginia*, an unpublished manuscript consisting of 168 watercolor ____ a detailed portrait of a particular species, such as *Amanita jacksonii*, a fungi species of the order Agaricales presented on plate 124.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) plates, each

B) plates, each was

C) plates; each being

D) plates; each

22

Every US state has an associated state soil, which is typically selected by a group of experts, then passed through the state legislature to receive its official designation. For example, Florida's Myakka soil was formally designated in 1989, and Nebraska's Holdrege soil in 1979. ____ years pass between a soil's selection and official designation, as the legislative process can be notoriously slow.

Which choice completes the text with the most logical transition?

- A) Therefore,
- B) Indeed,
- C) Similarly,
- D) In many cases,

23

In a 1984 study by Menon and Price, the researchers' aim was to analyze the diet composition of wild goats in Corsica, France. _____ they aimed to analyze the ratio of three different plant subtypes within these animals' diet: graminoids, forbs, and browse.

Which choice completes the text with the most logical transition?

- A) All the same,
- B) Therefore,
- C) Specifically,
- D) Instead,

24

Some former Olympic demonstration events, like taekwondo, were used to gauge the sport's popularity for future inclusion in the official Games, while others showcased lesser-known sports popular in that year's host country. _____ sumō, a Japanese form of wrestling, may have lacked international name recognition, but it was an apt demonstration sport at the 1964 Summer Olympic Games in Tokyo, Japan.

Which choice completes the text with the most logical transition?

- A) Conversely,
- B) Indeed,
- C) Similarly,
- D) Moreover,

25

While researching a topic, a student has taken the following notes:

- Elena Redbird was a classical composer and citizen of the Cherokee Nation.
- Her compositions synthesize Western classical music with elements of various Native musical traditions.
- Redbird's classical piece Songs for the River features a Lakota hymn.
- Redbird's classical piece Moonlit Cedar incorporates a Tlingit box drum, a traditional Native instrument.

The student wants to emphasize a similarity between the two classical compositions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Redbird's composition Songs for the River features a Lakota hymn, while Moonlit Cedar incorporates a Tlingit box drum.
- B) Redbird's works synthesize Western classical music and elements of various Native musical traditions; for example, the classical piece Songs for the River features a Lakota hymn.

C) Redbird's compositions synthesize Western classical music with elements of various Native musical traditions, such as a Lakota hymn.

D) Redbird's Songs for the River features a Lakota hymn, and Moonlit Cedar incorporates a Tlingit box drum; as such, both synthesize Western classical music with elements of Native musical traditions.

26

While researching a topic, a student has taken the following notes:

- The World Food Prize is awarded annually for contributions toward improving the world food supply.
- Lucia Moreno is a Peruvian food scientist who was awarded the prize in 2018.
- Moreno was awarded the prize with three researchers who worked alongside her.
- Moreno's group developed biofortified breeds of cassava plants.
- Kwame Mensah is a Ghanaian economist who was awarded the prize in 2019.
- He spearheaded economic initiatives promoting African agricultural infrastructure.

The student wants to emphasize a similarity between Moreno and Mensah. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A Peruvian food scientist, Moreno won the World Food Prize in 2018; in 2019, however, the annual prize was awarded to a Ghanaian economist—Mensah.
- B) Moreno won the World Food Prize in 2018 for her work with biofortified cassava plants, as did Mensah, a Ghanaian economist.
- C) Moreno, a food scientist and World Food Prize winner, has made scientific advancements with cassava plants, while Mensah, an economist, has promoted African agricultural infrastructure.
- D) Though one is a food scientist and the other an economist, Moreno and Mensah have both been recognized with World Food Prizes for contributions toward improving the world food supply.

27

- Leila Morgan (1912–1980) was an actress born in Chennai (then known as Madras), India.
- She was of Indian, Maori, and Irish heritage.
- She was the first Indian-born actress to be nominated for an Academy Award.
- Early in her career, she played many nameless, uncredited roles, such as her role in Harbor Lights (1931).
- Later, she played many named, credited roles, such as Clara Venn in A Day in a Woman's Life (1953).

Which choice most effectively uses information from the given sentences to specify Leila Morgan's role in A Day in a Woman's Life (1953)?

- A) In 1953, Indian-born actress Leila Morgan played the role of Clara Venn in the film A Day in a Woman's Life.
- B) Though she would go on to receive credit in many films, Leila Morgan started her career playing uncredited, nameless roles in films such as Harbor Lights (1931).
- C) Leila Morgan's story begins in Chennai (then known as Madras), India, in 1912.

D) Leila Morgan was an Indian-born actress who performed in many films, including Harbor Lights (1931) and A Day in a Woman's Life (1953).

Math Module 1

22 QUESTIONS

1

Maya bought stickers and granola bars for \$ 145. Each package of stickers cost \$ 3, and each granola bar cost \$ 1. If Maya bought 25 packages of stickers, how many granola bars did she buy?

- A) 145
- B) 75
- C) 70
- D) 120

2

If $7 + x = 1$, what is the value of $-22 - 2x$?

- A) 12
- B) -10
- C) -4
- D) 34

3

The length of each side of a square is 68 inches. Which expression gives the area of the square?

- A) $68/68$ square inches
- B) $68 + 68$ square inches
- C) $68 * 68$ square inches
- D) 18068 square inches

4

The measures of the interior angles of triangle ABC are 34° , 58° , and x° . What is the value of x ?

- A) 92
- B) 180
- C) 88
- D) 34

5

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(9)$?

- A) 27
- B) -72

C) 35

D) 19

6

$$7x - 4y = 5$$

$$2x + y = 7$$

The solution to the given system of equations is x, y . What is the value of $x + y$?

7

Of the 180 soccer players at a soccer camp, 72 are midfielders. What percentage of the soccer players at the camp are midfielders?

A) 72%

B) 40%

C) 7.2%

D) 0.4%

8

Maya is enlarging the area of a photograph that measures 7 centimeters by 5 centimeters. The resulting area, in square centimeters, of the photograph is given by the function $S(x) = (7x)(5x)$, where x represents the scale factor used to increase both the length and width of the photograph. Which statement is the best interpretation of $S(3) = 315$?

A) When the scale factor used to enlarge the photograph is 3, the resulting area is 7 square centimeters.

B) When the scale factor used to enlarge the photograph is 15, the resulting area is 35 square centimeters.

C) When the scale factor used to enlarge the photograph is 3, the resulting area is 315 square centimeters.

D) When the scale factor used to enlarge the photograph is 315, the resulting area is 3 square centimeters.

9

Which expression is equivalent to $7x^3 + 4x - 5 - 9x^2 + 6x + 3$?

A) $7x^3 - 9x^2 + 10x - 2$

B) $7x^3 - 9x^2 - 2x - 8$

C) $-2x^3 + 2x - 8$

D) $-2x^3 + 10x - 2$

10

points A and B lie on a circle with center C. The length of segment AC is 86 millimeters. What is the length, in millimeters, of segment BC?

- A) 172
- B) 43
- C) 86
- D) 344

11

$$49x^2 - 18 = 7$$

Which of the following is a solution to the given equation?

- A) $25/49$
- B) $5/49$
- C) $5/7$
- D) $11/49$

12

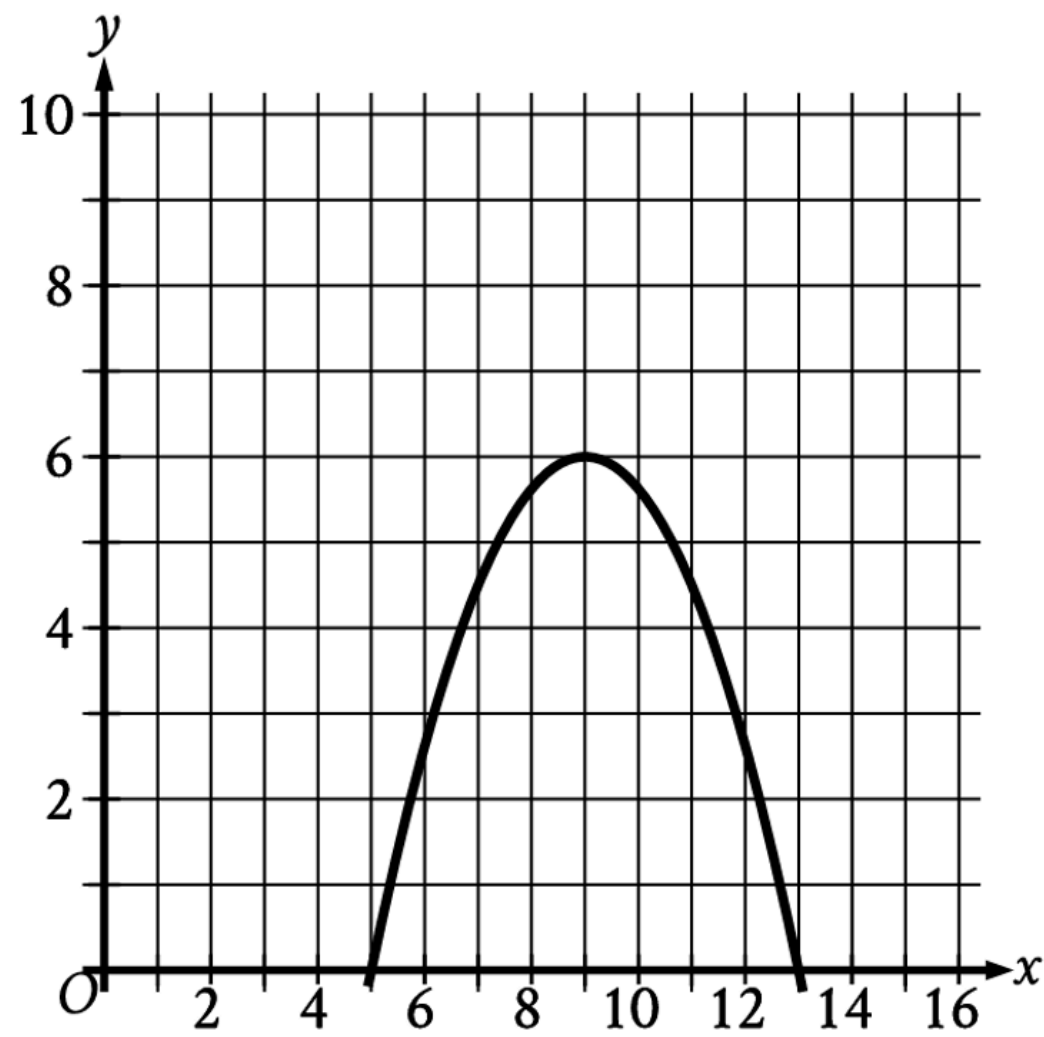
The height of a right circular cylinder is 47 inches, and the circumference of its base is 376 inches. Which expression represents the total surface area, in square inches, of the cylinder?

- A) $17672 + \pi 188^2/\pi$
- B) $\pi 188^2/\pi *47$
- C) $17672 + 2(188^2/\pi)$
- D) 17672

13

In the xy-plane, line k has a slope of 4 and a y-intercept of (0,-28). What is the x coordinate of the x-intercept of line k?

14



↩

An equation of the graph shown is $y = (-3/8)(x - p)^2 + 6$, where p is an integer constant. What is the value of p ?↩

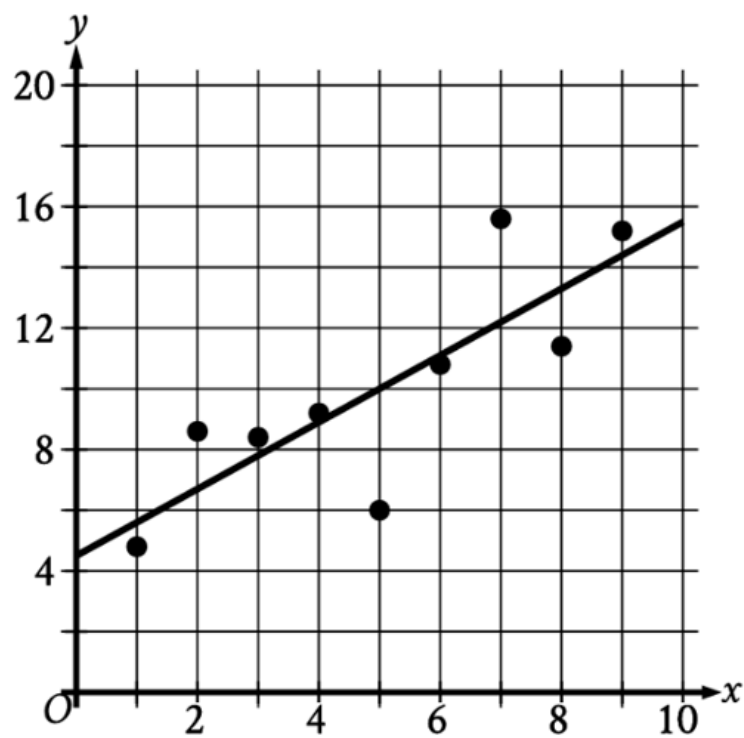
15

To cut a lawn, Jamal charges a fee of \$ 18.00 for his equipment and \$ 8.75 per hour spent cutting a lawn. Sofia charges a fee of \$ 10.00 for her equipment and \$ 10.75 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are all the values of x for which Sofia's total charge is greater than Jamal's total charge?

- A) $x < 3$
- B) $4 \leq x \leq 5$
- C) $x > 4$
- D) $3 \leq x \leq 4$

16

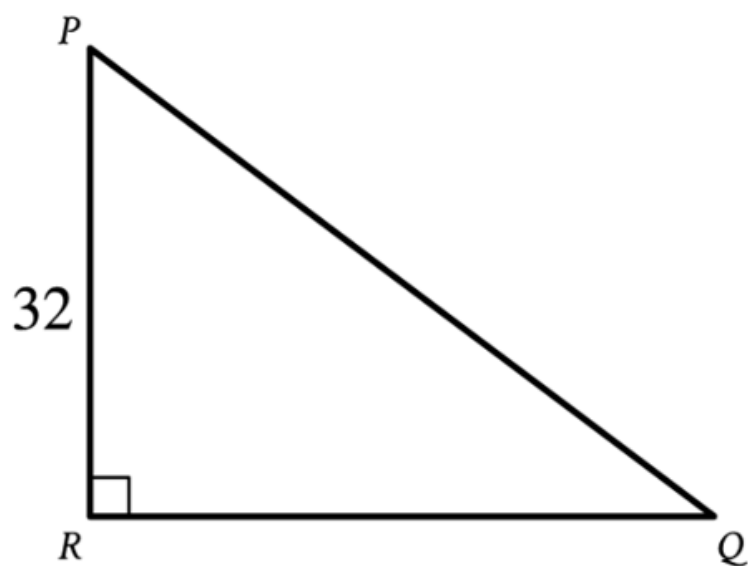
The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



At $x = 5$, how many units greater is the y -value predicted by the line of best fit than the actual y -value of the data point?

- A) 2
- B) 4
- C) 6
- D) 10

17



Note: Figure not drawn to scale.

In triangle PQR , the sine of $\angle Q$ is $\frac{1}{2}$. What is the length of $\overline{PQ}$?

18

For an online trivia game, 300 points are awarded for a correct answer if a question is answered in less than 4 seconds from the time the question is asked. Each 4 seconds after the question is asked, the number of points awarded for a correct answer decreases by 15% of the number of points

awarded for a correct answer in the previous 4 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 4?

- A) $f(x) = 300 * (0.15)^{x/4}$
- B) $f(x) = 300 * (0.85)^{4x}$
- C) $f(x) = 300 * (0.85)^{x/4}$
- D) $f(x) = 300 * (0.15)^{4x}$

19

$$5x + 4y = a$$

$$6x + 5y = b$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy -plane intersect at the point $(x, 4)$. What is the value of $5b - 6a$?

- A) -4
- B) 10
- C) 4
- D) -5

20

The function g is defined by $g(x) = (x + 17)(t - x)$, where t is a constant. In the xy -plane, the graph of $y = g(x)$ passes through the point $(24, 0)$. What is the value of $g(0)$?

21

$$t(v - 49) + t = w$$

The given equation relates the positive numbers w , t , and v . Which equation correctly expresses v in terms of t and w ?

- A) $v = \frac{w - t}{t} + 48$
- B) $v = \frac{w}{t} - 49$
- C) $v = \frac{w}{t} + 48$
- D) $v = w - t - 49$

22

x	gx
-20	2
-10	-3
1	8

The table shows three values of x and their corresponding values of gx , where $gx = \frac{fx}{x+6}$ and f is a linear function. What is the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane?

Math Module 2

22 QUESTIONS

1

The table shows the frequency of each data value in a data set.

Data value	Frequency
12	9
24	27
36	16
48	13

Which data value has a frequency of 27 in this data set?

- A) 36
- B) 12
- C) 48
- D) 24

2

$3 \mid x \mid =45$

Which value is a solution to the given equation?

- A) 42
- B) 15
- C) 48

D) 135

3

A bakery buys bags for \$ 0.06 each and boxes for \$ 0.38 each. The total cost of x bags and y boxes is \$ 524.72. Which equation represents this situation?

- A) $0.38x - 0.06y = 524.72$
- B) $0.38x + 0.06y = 524.72$
- C) $0.06x + 0.38y = 524.72$
- D) $0.06x - 0.38y = 524.72$

4

$$(p+7) + 6 = 11$$

What value of p is the solution to the given equation?

- A) 13
- B) -2
- C) 24
- D) 5

5

A train is traveling at a constant speed along a straight portion of railway. The equation $d = 40t$ gives the distance d , in feet from a railway marker, that the train will be t seconds after passing the marker. How many feet from the marker will the train be 6 seconds after passing the marker?

- A) 46
- B) 240
- C) 40
- D) 280

6

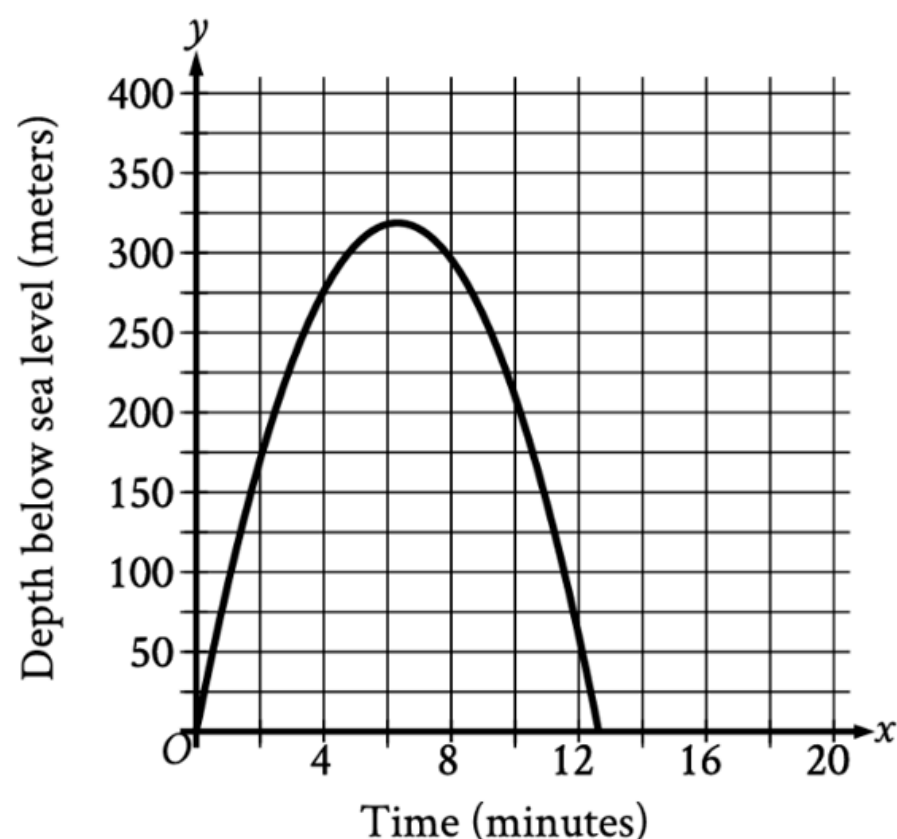
At a particular restaurant, the total cost of 6 chicken wings and a salad is 15.30 dollars. If the cost of each chicken wing is 1.25 dollars and there is no sales tax, what is the cost, in dollars, of the salad?

- A) 7.50
- B) 14.05
- C) 1.25
- D) 7.80

7

A certain salmon shark can swim at a maximum speed of 16 meters per second. At this maximum speed, how many meters would this salmon shark swim in 12 seconds?

8



The graph shows the predicted depth below sea level y , in meters, of a certain seal x minutes after the seal dives into the water. Based on the graph, which of the following is closest to the predicted depth below sea level, in meters, of the seal 2 minutes after the seal dives into the water?

- A) 0
- B) 170
- C) 276
- D) 318

9

Which expression is equivalent to $240x^2 + 56$?

- A) $308x^2 + 1$
- B) $78x^2 + 30$
- C) $8(30x^2 + 7)$
- D) $296(x^2 + 1)$

10

If $5 - 3x = 48$, what is the value of $(1/6)(5 - 3x)$?

- A) 48
- B) 6
- C) 8
- D) 288

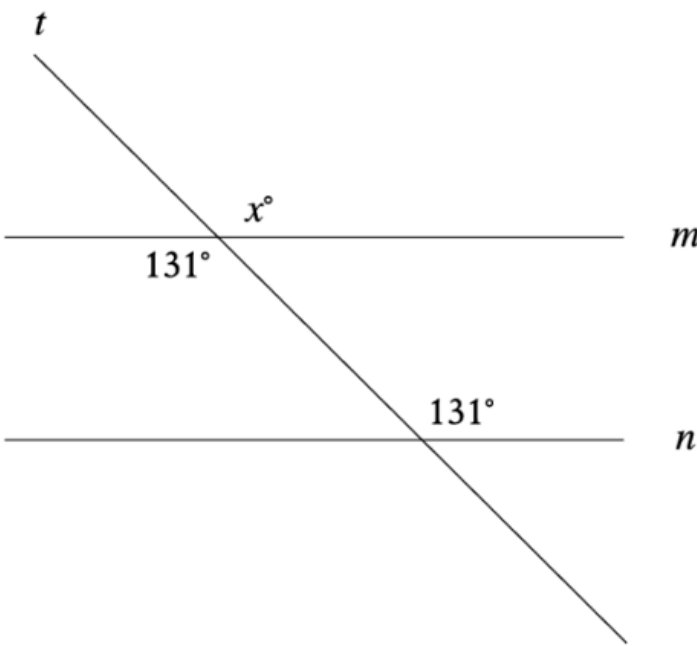
11

In the xy -plane, the graph of $y = 1/x$ is shifted up 24 units. Which of the following is an equation of the resulting graph?

- A) $y = 1/(x+24)$

- B) $y = 1/x - 24$
- C) $y = 1/(x-24)$
- D) $y = 1/x + 24$

12



Note: Figure not drawn to scale.

In the figure, lines m and n are parallel and line t intersects both lines. What is the value of x ?

13

$x > 55$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
50	0
51	0
52	0

B)

x	y
56	0
57	0
58	0

C)

x	y
45	0
46	0
47	0

D)

x	y
35	0
36	0
37	0

14

A local deli sells three types of sandwiches: turkey, ham, and veggie. The table shows the number of sandwiches of each type that the deli has available to sell, where a , b , and c are distinct positive integer constants.

Turkey	a
Ham	b
Veggie	c
Total	156

If a sandwich is selected at random, which expression represents the probability of selecting a sandwich that is **not** ham?

- A) $(a+c) / 156$
- B) $(a+b+c) / 156$
- C) $(b+c) / 156$
- D) $(a+b) / 156$

15

The function h is defined by $h(x) = 8 / (2x + 3)$. What is the value of $h(4)$?

16

Line j is defined by $y = 4x - 5$. Line k is parallel to line j in the xy -plane. What is the slope of line k ?

17

$$f(x) = 12x^{10} + 5x^8$$

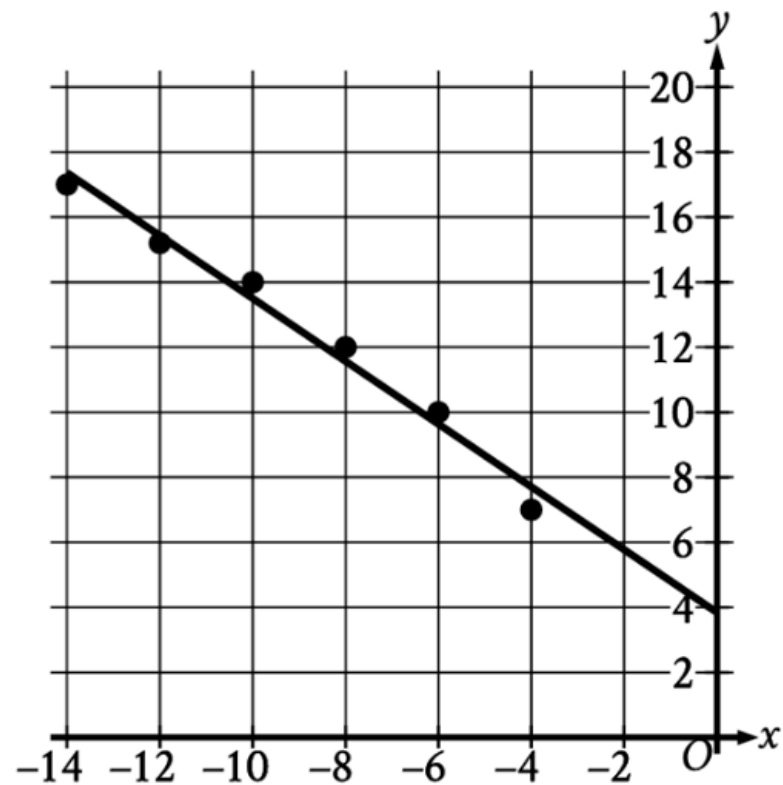
$$g(x) = -14x^7 + 9x^5$$

←

The polynomial $p(x)$ is defined as the product of the given polynomials, $f(x)$ and $g(x)$. What is the coefficient of x^{15} in $p(x)$?←

- A) 108
- B) 38
- C) 21
- D) 70

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A) $y = x - 17.4$
- B) $y = -x + 17.4$
- C) $y = x - 3.8$
- D) $y = -x + 3.8$

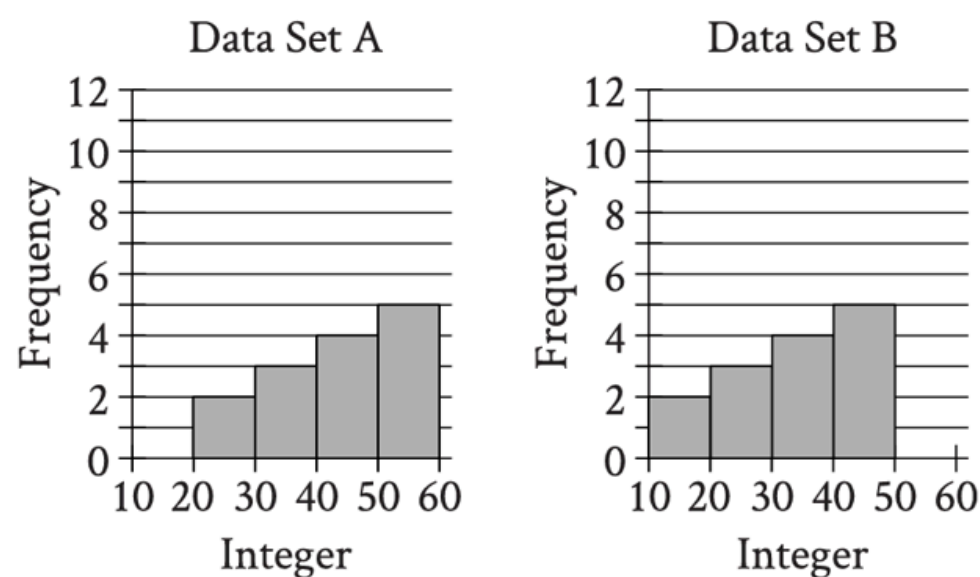
Circle K has a radius of 4 millimeters (mm). Circle L has an area of 81π mm². What is the total area, in mm², of circles K and L?

- A) 26π
- B) 97π
- C) 13π
- D) 52π

The population of a certain city doubled every 60 years from 1742 to 1982. The population of this city was 320,000 in 1982. What was the population of this city in 1742?

A scientist studied the effects of mixing two chemicals, chemical A and chemical B, by gradually adding more of each chemical to a mixture of the two chemicals. At the start of the study, the mass of chemical A in the mixture was equal to the mass of chemical B in the mixture. From the start of the study to the end of the study, the mass of chemical A in the mixture increased by 2,400% and the

- mass of chemical B in the mixture increased by 280%. At the end of the study, approximately how many times greater was the mass of chemical A in the mixture than the mass of chemical B in the mixture?
- A) 21.20
 - B) 6.58
 - C) 26.80
 - D) 8.57



- Two data sets of 14 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?
- A) 0
 - B) 1
 - C) 10
 - D) 14

Reading and Writing Module 1 Answers

- 1. A
- 2. A
- 3. A
- 4. A
- 5. A
- 6. D
- 7. B
- 8. D
- 9. C
- 10. D
- 11. A
- 12. A

- 13. D
- 14. D
- 15. D
- 16. A
- 17. B
- 18. D
- 19. D
- 20. A
- 21. C
- 22. B
- 23. B
- 24. B
- 25. D
- 26. B
- 27. D

Reading and Writing Module 2 Answers

- 1. A
- 2. A
- 3. C
- 4. D
- 5. B
- 6. D
- 7. B
- 8. B
- 9. A
- 10. A
- 11. B
- 12. D
- 13. B
- 14. B
- 15. D
- 16. D
- 17. D
- 18. B
- 19. A
- 20. A
- 21. A
- 22. D
- 23. C
- 24. B
- 25. D
- 26. D
- 27. A

Math Module 1 Answers

1. C
2. B
3. C
4. C
5. D
6. 4.8
7. B
8. C
9. A
10. C
11. C
12. C
13. 7
14. 9
15. C
16. B
17. 64
18. C
19. C
20. 408
21. C
22. 52

Math Module 2 Answers

1. D
2. B
3. C
4. B
5. B
6. D
7. 192
8. B
9. C
10. C
11. D
12. 131
13. B
14. A
15. $\frac{8}{11}$
16. 4
17. B
18. D

19. B

20. 20000

21. B

22. B